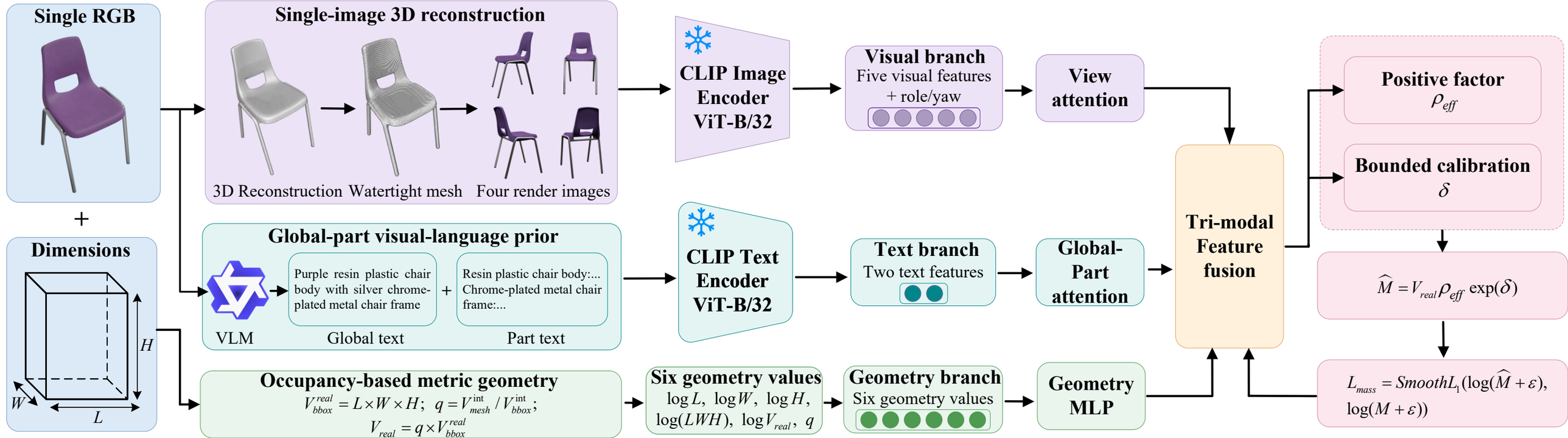

Single RGB
+
Dimensions
H
W
L
Single-image 3D reconstruction
3D Reconstruction
Watertight mesh
Four render images
Global-part visual-language prior
VLM
Purple resin plastic chair body with silver chrome-plated metal chair frame
Global text
+
Resin plastic chair body:... Chrome-plated metal chair frame:...
Part text
Occupancy-based metric geometry
$V_{bbox}^{real} = L \times W \times H;\ q = V_{mesh}^{int} / V_{bbox}^{int};$
$V_{real} = q \times V_{bbox}^{real}$
CLIP Image Encoder ViT-B/32
CLIP Text Encoder ViT-B/32
Visual branch
Five visual features + role/yaw
Text branch
Two text features
Six geometry values
$\log L,\ \log W,\ \log H,$
$\log(LWH),\ \log V_{real},\ q$
Geometry branch
Six geometry values
View attention
Global-Part attention
Geometry MLP
Tri-modal Feature fusion
Positive factor
ρ_{eff}
Bounded calibration
δ
$\hat{M} = V_{real}\rho_{eff}\exp(\delta)$
$L_{mass} = SmoothL_1(\log(\hat{M}+\varepsilon), \log(M+\varepsilon))$